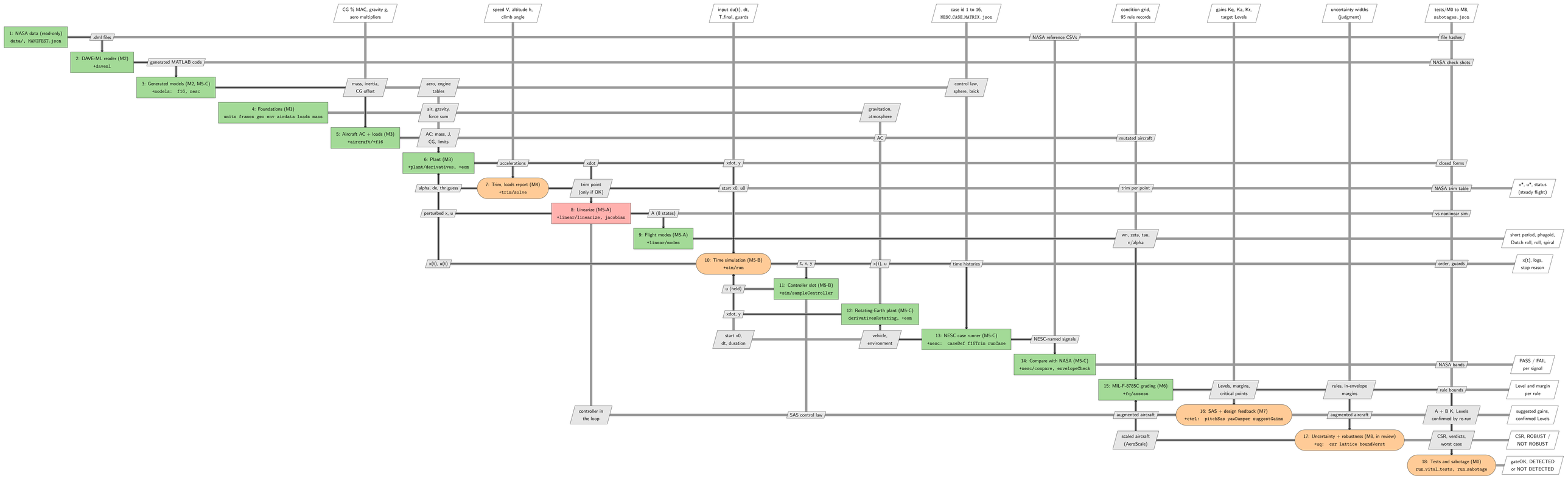
CG % MAC, gravity g, aero multipliers
speed V, altitude h, climb angle
input du(t), dt, T_final, guards
case id 1 to 16, NESC.CASE_MATRIX.jsonn
condition grid, 95 rule records
gains Kq, Ka, Kr, target Levels
uncertainty widths (judgment)
tests/M0 to M8, sabotages.json
1: NASA data (read-only) data/, MANIFEST.json
.dml files
NASA reference CSVs
file hashes
2: DAVE-ML reader (M2) +daveml
generated MATLAB code
NASA check shots
3: Generated models (M2, M5-C) +models: f16, nesc
mass, inertia, CG offset
aero, engine tables
control law, sphere, brick
4: Foundations (M1) units frames geo env airdata loads mass
air, gravity, force sum
gravitation, atmosphere
5: Aircraft AC + loads (M3) +aircraft/+f16
AC: mass, J, CG, limits
AC
mutated aircraft
6: Plant (M3) +plant/derivatives, +eom
accelerations
xdot
xdot, y
closed forms
alpha, de, thr guess
7: Trim, loads report (M4) +trim/solve
trim point (only if OK)
start x0, u0
trim per point
NASA trim table
x*, u*, status (steady flight)
perturbed x, u
8: Linearize (M5-A) +linear/linearize, jacobian
A (8 states)
vs nonlinear sim
9: Flight modes (M5-A) +linear/modes
wn, zeta, tau, n/alpha
short period, phugoid, Dutch roll, roll, spiral
x(t), u(t)
10: Time simulation (M5-B) +sim/run
t, x, y
x(t), u
time histories
order, guards
x(t), logs, stop reason
u (held)
11: Controller slot (M5-B) +sim/sampleController
xdot, y
12: Rotating-Earth plant (M5-C) derivativesRotating, +eom
start x0, dt, duration
vehicle, environment
13: NESC case runner (M5-C) +nesc: caseDef f16Trim runCase
NESC-named signals
14: Compare with NASA (M5-C) +nesc/compare, envelopeCheck
NASA bands
PASS / FAIL per signal
15: MIL-F-8785C grading (M6) +fq/assess
Levels, margins, critical points
rules, in-envelope margins
rule bounds
Level and margin per rule
controller in the loop
SAS control law
augmented aircraft
16: SAS + design feedback (M7) +ctrl: pitchSas yawDamper suggestGains
augmented aircraft
A + B K, Levels confirmed by re-run
suggested gains, confirmed Levels
scaled aircraft (AeroScale)
17: Uncertainty + robustness (M8, in review) +uq: csr lattice boundWorst
CSR, verdicts, worst case
CSR, ROBUST / NOT ROBUST
18: Tests and sabotage (M0) run.vital.tests, run.sabotage
gateOK, DETECTED or NOT DETECTED